

CROSS-TRIAD RIGIDITY FOR CASCADE OPERATORS

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ABSTRACT. Paper II shows that on any single triad a sum of cascade terms reproduces an Euler–Arnold vertex, so what remains of the sum question is whether one fixed coefficient set can do so on *all* triads at once. For a two-species cascade this is a gauge-trivialisation problem, and we settle it negatively for every scalar inertia multiplier. A first invariant — the ratios of the blocks’ Cayley hyperdeterminants — excludes all but two exponents, $s = 0$ and $s = -2$, which are Euler and its vorticity dual. We show that this degeneracy is structural, by proving that on the relevant stratum the hyperdeterminant reduces to the product of the block’s four entries, and that it is nevertheless only a weakness of that invariant: the correct condition, that all the vertices lie in a single $GL(2)^3$ orbit, fails at $s = 0$ and $s = -2$ as well.

1. THE QUESTION

By Paper II, Theorem 5.7 every Euler–Arnold vertex on a triad is a sum of at most eight tensors with one rank-one block each, so no triad-local argument can exclude sums. What that construction does not produce is a cascade operator in Tao’s sense: its terms are built from the vertex itself and change from triad to triad, whereas a local cascade operator reuses one finite coefficient set on every triad and at every scale. The contribution of a cascade with m species to a triad $k + l + m = 0$ is

$$\sum_{i_1, i_2, i_3} \alpha_{i_1 i_2 i_3} \hat{\psi}_{i_1}(k) \otimes \hat{\psi}_{i_2}(l) \otimes \hat{\psi}_{i_3}(m),$$

the α constant, the divergence-free profiles $\hat{\psi}_i(k) \in k^\perp$ varying with the wavevector.

Question 1.1. *Do there exist m , profiles, constants α , and an inertia multiplier A with this equal to $T_A(k, l, m)$ on every triad of the support?*

For $m = 2$ the profiles form a frame G_k of each perp-plane and the question becomes a trivialisation,

$$T_A(k, l, m) = (G_k \otimes G_l \otimes G_m) \alpha, \quad \alpha \text{ independent of the triad}, \quad (1)$$

equivalently: all the vertices lie in one orbit of $GL(2)^3$ acting on $\mathbb{R}^{24}$.

2. A FIRST INVARIANT, AND WHERE IT FAILS

Proposition 2.1. *If (1) holds then the ratios $\text{Det}(T_0) : \text{Det}(T_1) : \text{Det}(T_2)$ of the blocks’ Cayley hyperdeterminants are the same on every triad.*

Proof. $\text{Det}((g_1 \otimes g_2 \otimes g_3)t) = (\det g_1 \det g_2 \det g_3)^2 \text{Det}(t)$, and block j carries its slots on the legs $j, j + 1, j + 2$, that is on all three legs. So every block is acted on by the same triple and each $\text{Det}(T_j)$ picks up the same factor. $\square$

Proposition 2.2 (numerical). *The ratios are not triad-independent. For $A = |k|^s I$ with $s \notin \{0, -2\}$, and for the anisotropic multiplier $A_k = \text{diag}(1, 1 + |k|)$, the hyperdeterminants are of order 10^{-3} to 10^{-2} relative to $\|T\|^4$ and $\text{Det}(T_0)/\text{Det}(T_2)$ varies: for $s = -3$ it takes the values 0.418, 0.827, 1.101, 0.743, 1.507 at radii $(1, 1.1, 1.2)$, $(1, 1.3, 1.6)$, $(1, 1.5, 1.8)$, $(1, 1.2, 2.0)$, $(1, 1.9, 2.4)$. Exactly two exponents escape: at $s = 0$ and $s = -2$, and for every constant $A = cI$, all three hyperdeterminants vanish identically, to 10^{-17} relative.*

3. THE DEGENERACY IS STRUCTURAL

For scalar A the operators P_i, Q_i are scalars, so a block is supported on the four monomials of $S^A \cup S^B$. Write its entries

$$p = t_{111}, \quad q = t_{122}, \quad r = t_{212}, \quad w = t_{221}.$$

Lemma 3.1. *On this stratum $\text{Det}(t) = 4pqrw$.*

Proof. In Cayley's expression every monomial contains one of $t_{112}, t_{121}, t_{211}, t_{222}$ except the single term $4t_{111}t_{122}t_{212}t_{221}$. $\square$

Theorem 3.2. *Let $A = f(|k|)I$. Then all three hyperdeterminants vanish on every triad if and only if f is constant or $f(x) = c/x^2$; that is, $s = 0$ or $s = -2$.*

Proof. With $c^A = u_l/(u_k u_m)$ and $c^B = u_k/(u_l u_m)$, $u_i = \mu_i^{1/2}$, and the coefficients of Paper I, Lemma 5.7,

$$q = \frac{1}{2}c^A\gamma, \quad r = \frac{1}{2}c^B\gamma', \quad w = \frac{J}{2z}(c^B - c^A), \quad p = \frac{J}{2z}\left(c^A\frac{y}{x} - c^B\frac{x}{y}\right).$$

Since $\gamma = -J/x$ and $\gamma' = J/y$ are non-zero on a non-degenerate triad, q and r never vanish, so by Lemma 3.1 the hyperdeterminant vanishes exactly when $p = 0$ or $w = 0$. Now $c^A/c^B = \mu_l/\mu_k$, so $w = 0$ on all triads iff $f(y) = f(x)$ for all admissible x, y , i.e. f is constant; and $p = 0$ on all triads iff $f(y)y^2 = f(x)x^2$, i.e. $f(x)x^2$ is constant. $\square$

So the two exceptions are not an accident of the sample: they are exactly the two inertia operators for which one of the four entries of every block vanishes identically. The two are exchanged by a duality fixed at $s = -1$, where the block is self-dual, its entries satisfying $p = w$ and $q = -r$.

4. BUT THE INVARIANT'S SILENCE IS AN ACCIDENT

Lemma 3.1 also shows why Proposition 2.1 is weak here: on this stratum the hyperdeterminant is merely a product of coordinates, and it reports nothing beyond whether all four are non-zero. The natural next invariants are no better. Pairing two blocks with the alternating form, $\langle X, Y \rangle_\varepsilon := \varepsilon^{aA} \varepsilon^{bB} \varepsilon^{cC} X_{abc} Y_{ABC}$ after aligning both by leg, is a relative invariant of weight one in each $\det G$, so ratios of such pairings would be absolute invariants. They vanish identically: the four monomials above are exactly those of even parity, ε couples even parity to odd, and so every such pairing is zero. This is the parity grading of Paper I, Lemma 5.9 reappearing.

One should therefore test the condition itself rather than a proxy.

Proposition 4.1 (numerical). *Let $A = f(|k|)I$ be a scalar multiplier. At a triad let V be the span of the twelve orbit directions $E \cdot T$, $E \in \mathfrak{gl}(2)^3$, together with the three directions $\partial T/\partial \mu_i$ — the latter admissible because f is a free function and only its three values at the leg lengths enter. Then the shape derivatives $\partial T/\partial y$, $\partial T/\partial z$ do not lie in V . Their relative residuals after projection onto V , at radii $(1, 1.5, 1.8)$, are 0.065 and 0.055 for $s = 0$; 0.39 and 0.54 for $s = -2$; and between 0.15 and 0.35 for $s = -1$, $s = 3/2$ and $s = 3$.*

Corollary 4.2. *For every scalar inertia multiplier, including Euler and the vorticity form, Question 1.1 has a negative answer with $m = 2$.*

Remark 4.3. Residuals are reported rather than ranks. A rank count needs a tolerance, and at radii (1, 1.3, 1.6) the Euler residual is only 2.4×10^{-5} : non-zero, and far above the $O(h^2) \approx 10^{-10}$ truncation of the central differences used, but small enough that a fixed-tolerance rank is unreliable there. The escape at $s = 0$ is genuine but weak at that shape, and plain at others.

Remark 4.4. The computation was checked three ways: the orbit directions lie in the kernel of the energy relations of Paper II, Theorem 5.1 to 5×10^{-17} , as they must since the gauge action preserves $\langle T(u, u), u \rangle$; a finite group motion lies in their span to the accuracy of the difference quotient; and the shape derivative lies in that kernel too.

5. THE SINGLE-TRIAD METHOD IS EXHAUSTED

Proposition 4.1 does not extend to matrix-valued multipliers, and the reason is not that the test was run too coarsely.

Proposition 5.1 (numerical). *Let A_k be a full positive definite operator on $k^\perp$, so that the admissible span is the twelve orbit directions together with the nine directions $\partial T / \partial A_i$. Then the shape derivatives lie in that span, to relative residual below 10^{-11} , for every inertia operator tested — scalar, diagonal anisotropic, fully anisotropic, and 200 random positive definite triples — and at both triads. The span has dimension 14 inside the 16-dimensional $\ker L$, so this is not a case of the admissible directions exhausting the ambient space.*

So with three functions per wavevector instead of one, the gauge freedom and the multiplier together absorb the shape variation of the vertex exactly, to first order. On a single triad there is nothing left to obstruct, and no refinement of the present argument will recover one.

6. THE WEB

In both cases what defeats the argument is that a single triad sees each leg once. The information not used is that G_k and A_k are functions of the wavevector, shared by every triad through k . For $m = 2$ that sharing yields a condition in closed form, and one that holds for a matrix multiplier as well.

Lemma 6.1. *Suppose $X = (G_k \otimes G_l \otimes G_m)\alpha$ with $m = 2$. Then for $v \in k^\perp$ the slice $X(v) := X(v, \cdot, \cdot)$ satisfies $X(v) = G_l \alpha(G_k^\top v) G_m^\top$, so*

$$\det X(v) = \det G_l \det G_m \cdot q_\alpha(G_k^\top v), \quad q_\alpha(w) := \det \alpha(w, \cdot, \cdot).$$

The triad enters only through a scalar. Hence for any two triads τ, τ' sharing the leg k , the binary quadratic forms $Q_\tau(v) := \det T_A(\tau)|_v$ and $Q_{\tau'}$ on $k^\perp$ are proportional.

This is the first necessary condition available for a matrix-valued multiplier, Proposition 5.1 having closed the single-triad route. We tested it on a web: legs the 28 non-zero points of a planar lattice ball of radius 3, triads the 60 non-degenerate triples summing to zero, giving 395 pairs of triads with a common leg.

Proposition 6.2 (numerical). *The condition excludes generic multipliers. The proportion of shared-leg pairs passing is 0% for independent random positive definite A_k , with minimum separation 1.7×10^{-3} over 516 pairs; 15% for $A_k = \text{diag}(1, 1 + |k|)$; and 13% for $A = |k|^{3/2}I$. For $A = I$ and $A = |k|^{-2}I$ it holds exactly, on every pair.*

The naive count is encouraging and wrong. Proportionality of two binary quadratics is two conditions, so the web carries $\sum_k 2(n_k - 1) = 304$ of them against $3|S| = 84$ free parameters; conditions grow like $|S|^2$ and parameters like $|S|$. That does not survive contact with the computation.

Proposition 6.3 (numerical). *The conditions are massively redundant. Linearising the residual in the 84 parameters at $A = I$, the Jacobian has rank 26, so the local solution set is 58-dimensional; at $A = |k|^{-2}I$ the rank is 43 and the solution set is 41-dimensional. The web condition therefore does not isolate the two exceptional multipliers, and does not settle the matrix case.*

Proposition 6.4 (numerical). *The redundancy has the same source as every previous escape. At $A = I$ and $A = |k|^{-2}I$ each Q_τ has vanishing discriminant — to 0 and 5×10^{-15} respectively — so it is a perfect square ℓ_τ^2 ; and the line ℓ_τ is the same for every triad through a given leg, the directions agreeing to 3×10^{-16} . For $A = |k|^{3/2}I$ the discriminant is of order one and the directions spread maximally. Proportionality $\ell^2 \propto \ell'^2$ is therefore one condition rather than two, and it is satisfied automatically.*

Remark 6.5. That Q_τ is a perfect square is Theorem 3.2 again: the discriminant of the slice quadratic is the Cayley hyperdeterminant of the block, which vanishes identically exactly at $s = 0$ and $s = -2$. The three failures recorded in this paper — the hyperdeterminant ratios, the ε pairings, and now the web — are one failure, and it is that the blocks of the Euler and vorticity vertices are degenerate as $2 \times 2 \times 2$ tensors.

Remark 6.6. One positive fact emerges. For $A = I$ and $A = |k|^{-2}I$ every leg k carries a canonical line in $k^\perp$, the repeated root of Q_τ , and it does not depend on which triad through k is used to compute it. Any further attempt should either build on that line or explain why it cannot separate; it is the only triad-independent structure these two vertices have been found to carry.

7. THE CANONICAL LINES, AND WHY THEY DO NOT HELP

The last remark of Section 6 isolated the only triad-independent structure these two vertices carry. It can be identified, and it is richer than a single line.

Proposition 7.1 (numerical). *At $A = I$ every leg carries two distinct canonical lines: slicing a block in either of its input slots gives the plane normal ν , while slicing in the output slot gives $\hat{k}$. The discriminants vanish to 10^{-15} , and the two directions are exactly 90° apart, on every triad and at every leg. At $A = |k|^{-2}I$ all three slots give $\hat{k}$: a single line.*

If (1) holds, the lines at k are G_k applied to the corresponding canonical lines of α , which are fixed. So G_k is pinned to the stabiliser of the observed configuration: the diagonal torus, two parameters, at $A = I$; the Borel, three, at $A = |k|^{-2}I$; in place of the four of $GL(2)$. This is a real rigidification, and it is worth nothing.

Proposition 7.2 (numerical). *On a single triad the shape derivatives of T_A lie in the span of the nine derivatives $\partial T_A / \partial A_i$ alone, with no gauge directions at all. The relative residuals are below 3×10^{-10} for every inertia operator tested — $A = I$, $|k|^{-2}I$, $|k|^{3/2}I$, fully anisotropic — and at both triads, while that span has dimension only 8 or 9 inside the 16-dimensional $\ker L$. The statement is not vacuous: random vectors of $\ker L$ have residuals between 0.25 and 0.90 off the same span.*

Corollary 7.3. *Rigidifying the gauge cannot obstruct a matrix multiplier. Cutting the twelve gauge directions to the six allowed by Proposition 7.1 leaves the shape derivative absorbed to 1.4×10^{-12} , and cutting to the nine of the Borel case leaves it absorbed to 6×10^{-12} .*

Remark 7.4. The reason is now plain, and it is not about the gauge at all. Proposition 7.2 says that changing the shape of a triad is, to first order, the same as adjusting the inertia operator at its three legs. The multiplier freedom already covers the shape variation, so no restriction placed on G_k — by canonical lines or by anything else — can create an obstruction that a single triad is able to see. With A held fixed the picture is the opposite: the shape derivative then escapes the full twelve-dimensional gauge span with residuals between 0.39 and 0.71, which is Proposition 4.1 in another guise.

8. THE EXISTENCE QUESTION

A correction first. We said above that a further attempt must compare triads that do not share their multiplier data. That is backwards. Triads with disjoint legs carry independent A , so by Proposition 7.2 each absorbs its own shape and nothing is obstructed; rigidity can only come from triads that *do* share legs. That is what a web provides.

The question is also not undecidable. “There exist A , α and G such that $T_A(\tau) = (G \otimes G \otimes G)\alpha$ for every τ in a finite web” is a first-order sentence over the reals, and Tarski–Seidenberg decides such a sentence in principle. What is out of reach is the elimination, not the logic. So the question is one of measurement, and the quantity to measure is the dimension of two images inside $\mathbb{R}^{24|T|}$: that of $\{A_k\} \mapsto (T_A(\tau))_\tau$, and that of $(\alpha, \{G_k\}) \mapsto ((G \otimes G \otimes G)\alpha)_\tau$. A representation exists only if they meet.

Proposition 8.1 (numerical). *Measured by Jacobian rank at random points, on the planar lattice webs of radius 2 and 3:*

<i>web</i>	<i>ambient $24 T$</i>	<i>$m = 2$</i>	<i>$m = 3$</i>	<i>$m = 4$</i>
$R = 2$ (12 legs, 8 triads)	192	84	122	176
$R = 3$ (28 legs, 60 triads)	1440	198	268	354

the entries being $\dim V + \dim C$. The vertex image attains its parameter count exactly, $\dim V = 3|S|$; the cascade image falls short of $m^3 + 2m|S|$ only by the dimension of the gauge stabiliser. The codimension of the pair is 108, 70, 16 at $R = 2$ and 1242, 1172, 1086 at $R = 3$.

Both dimensions grow linearly in $|S|$ for every fixed m , while the ambient grows like $|T| \sim |S|^2$. The deficit therefore diverges, and it does so for Tao’s $m = 4$ exactly as for $m = 2$: the small web is nearly saturated at $m = 4$, with codimension 16, and the next one is not, with codimension 1086. This is the counting of Section 6 finally doing what was claimed for it there — and doing it because, unlike the proportionality conditions of Proposition 6.2, these images suffer no redundancy collapse.

Remark 8.2. What this does and does not establish. Two subvarieties whose dimensions sum to less than the ambient cannot meet in general position, but may still meet in special position, and a dimension count does not exclude that. So Proposition 8.1 is not a proof. It is, however, the strongest form the evidence can take short of elimination: the deficit is not marginal, it is not an artefact of redundancy, it grows without bound, and it is insensitive to the species count. On one triad the images do meet, trivially, since a single triad imposes no condition on α at all; the web is what separates them, and the separation widens as the web grows.

9. CONCLUSION

The sum question, as Paper II left it, is answered. Triad-locally, sums are not excluded: Paper II, Theorem 5.7 writes every Euler–Arnold vertex as a sum of at most eight admissible tensors. Globally, a local cascade operator cannot be an Euler–Arnold operator, and this paper says why at four levels of sharpness. No linear functional separates the classes, the only ones

vanishing on the Euler–Arnold class being the energy identity that every cascade satisfies by construction (Paper II, Corollary 5.4). No invariant of a single block separates them, the relevant blocks being degenerate as $2 \times 2 \times 2$ tensors (Theorem 3.2). No rigidification of the gauge separates them, since on one triad the inertia operator can imitate a change of shape (Proposition 7.2). And on a web the two classes are separated by a codimension that grows quadratically in the number of legs, for every species count (Proposition 8.1).

We stop here. The remaining gap is between a dimension count and an elimination, and closing it is a question in real algebraic geometry rather than in fluid mechanics.

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